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Simulating synchronization in neuronal networks

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We discuss several techniques used in simulating neuronal networks by exploring how a network's connectivity structure affects its propensity for synchronous spiking. Network connectivity is generated using the Watts-Strogatz small-world algorithm, and two key measures of network structure are described. These measures quantify structural characteristics that influence collective neuronal spiking, which is simulated using the leaky integrate-and-fire model. Simulations show that adding a small number of random connections to an otherwise lattice-like connectivity structure leads to a dramatic increase in neuronal synchronization. © *2016 American Association of Physics Teachers.*
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I. INTRODUCTION

Synchronous firing of networks of neurons is essential to information processing in the brain and has been associated with many cognitive processes, including directed attention,[1] memory encoding,[2] and binding of sensory features.[3] Hypersynchronous neuronal activity is also associated with brain pathologies such as epilepsy[4] and Parkinson's disease,[5] motivating the need to understand the various mechanisms by which neuronal networks synchronize. Experimental methods are currently limited in their ability to record large populations of neurons with simultaneously high spatial and temporal resolution,[6] making simulations of neuronal networks an excellent tool for addressing this problem.

The human brain is a highly coupled system, with each of its approximately 86×10^9 neurons[7] forming an average of $\sim 10^4$ connections with other neurons.[8] These neurons also act as oscillators, because neurons respond to injected current by firing at a fairly constant frequency. Thus, the brain features two essential ingredients (coupling and oscillation) known to facilitate synchronization in many physical systems, from pendulums to Josephson junctions to chaotic electrical circuits.[9–11] In networks of dynamical systems with many oscillators, the large-scale connectivity structure dramatically affects the ability of the system to synchronize.[12]

In this paper, we will explore the influence of network characteristics on neuronal synchronization by introducing a popular paradigm for generating network connectivity known as the Watts-Strogatz small-world network model. This simple but powerful model interpolates between nearest-neighbor connectivity and random connectivity by the change of a single parameter. Many experimental studies have demonstrated that neuronal networks exhibit some of this model's most salient characteristics.[13–15] We will also describe two important measures for characterizing network structure, explain how to compute them, and discuss their influence on network synchronization. Finally, to simulate the dynamics of individual neurons within the network, we will utilize the simplest biophysical neuron model, known as the leaky integrate-and-fire model, and explain a subtle but important point regarding its numerical integration.

II. THE LEAKY INTEGRATE-AND-FIRE MODEL

Due to its simplicity, the leaky integrate-and-fire model[16] is the ideal choice for focusing on the effects of connectivity structure in shaping neuronal network dynamics. The cell membrane, which at rest collects positively charged ions on its exterior surface and negatively charged ions on the interior, is modeled as a capacitor. The transmembrane proteins that permit current to flow across the cell membrane are collectively modeled as a resistor in parallel with the capacitor, along with a battery that models the typical neuron's resting membrane potential of approximately $-70\,\text{mV}$ (the potential inside the cell is 70 mV below the potential outside the cell). The corresponding circuit diagram is depicted in Fig. 1(a).

In contrast with more complex neuron models (such as Hodgkin-Huxley[17]) that incorporate active currents (associated with variable conductances/resistors) to model the time course of action potentials, the leaky integrate-and-fire model dispenses with any attempt to reproduce the rapid increase and decrease in membrane potential that constitute a spike. Instead, the passive dynamics associated with the fixed conductance g_L determine the evolution of the membrane potential until it breaches the threshold V_{thresh} at which the neuron undergoes a spike (usually around $-50\,\text{mV}$). At this point, the membrane potential is reset to a lower value (often the resting membrane potential). To model the typical duration of an action potential (1–2 ms), as well as the short time after an action potential during which the neuron is unable to fire, a *refractory period* (t_{refrac}) of several milliseconds is enforced,

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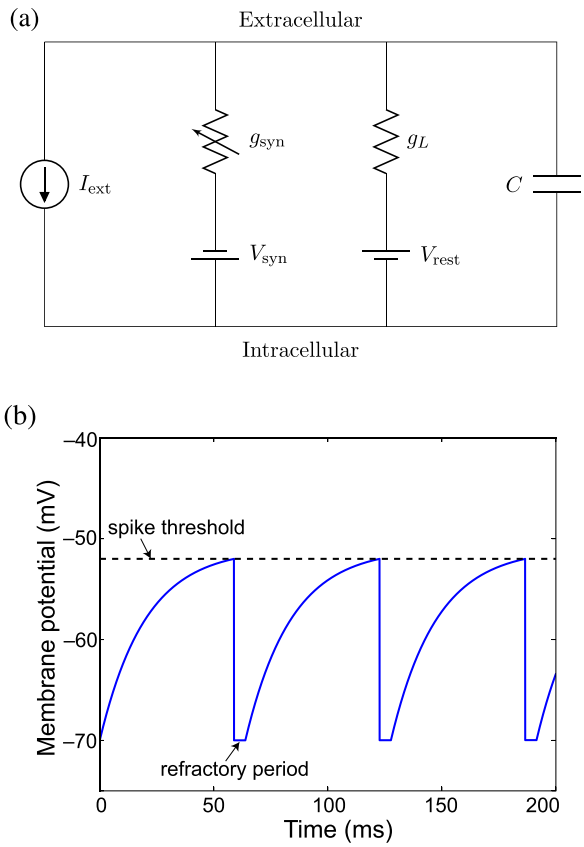

Fig. 1. Electrical circuit model and example voltage trace of the leaky integrate-and-fire model. (a) The voltage across the capacitor represents the membrane potential; V_{rest} is the resting membrane potential and g_L is the membrane conductance. The synaptic conductance g_{syn} increases when the neuron receives a signal (due to a pre-synaptic spike), resulting in the membrane potential being driven toward V_{syn}. I_{ext} represents the external current injected into the neuron. (b) Typical voltage trace showing how the membrane potential resets when it reaches V_{thresh}. The membrane potential is locked at the reset potential during the refractory period.

during which time the neuron's membrane potential is fixed at the reset value.

Experimentally, a stimulating electrode can inject current into a neuron to induce repetitive firing, a process that is modeled as a current source in parallel with the capacitor [far left branch in Fig. 1(a)]. Coupling with other neurons is modeled by a variable resistor and a battery (which reflects the ionic concentration gradients across the cell membrane) in parallel with the capacitor. If one neuron receives a connection from another neuron that fires an action potential, then the conductance of the recipient neuron's variable resistor, g_{syn}, undergoes a transient pulse. The voltage V_{syn} of the battery is typically in the vicinity of 0 mV for an excitatory connection, so that the current induced by the conductance pulse drives the membrane potential toward the firing threshold, potentially eliciting an action potential. Overall, the dynamics of an individual neuron's membrane voltage V are governed by

$$C\frac{dV(t)}{dt} = -g_L[V(t) - V_{rest}] + I_{syn}(t) + I_{ext}, \tag{1}$$

where $I_{syn}(t) = -g_{syn}(t)[V(t) - V_{syn}]$, with the understanding that V is discontinuously reset to V_{rest} when it breaches V_{thresh}.

This discontinuity must be treated carefully in network simulations of neurons in the leaky integrate-and-fire model. It has been shown that setting the membrane potential to exactly V_{rest} at the time step that V_{thresh} is breached can lead to inaccurate network dynamics.[18,19] Instead, it is important to recognize that if $V < V_{thresh}$ at time t and $V \geq V_{thresh}$ at time $t + \Delta t$, then t_{spike} (the exact time at which V_{thresh} is breached, and at which V should be reset to V_{rest}) lies sometime between t and $t + \Delta t$, so that the membrane potential at time $t + \Delta t$ is slightly greater than V_{rest}.

The simplest numerical solution is to calculate t_{spike} by linearly interpolating between the two times

$$t_{spike} = t + \left[\frac{V_{thresh} - V(t)}{V_0(t + \Delta t) - V(t)}\right]\Delta t, \tag{2}$$

where $V_0(t + \Delta t)$ is the membrane potential the neuron would attain after breaching V_{thresh}, were it not reset. This interpolation scheme can be shown[18] to lead to a reset potential of

$$V(t + \Delta t) = V_{rest} + [V_0(t + \Delta t) - V_{thresh}] \times \left[1 + \frac{\Delta t}{\tau}\frac{V(t) - V_{rest}}{V_0(t + \Delta t) - V(t)}\right], \tag{3}$$

where τ is the membrane time constant C/g_L. The combination of this technique with a second-order Runge-Kutta integration scheme (Heun's method[20]) reduces the relative error in the firing rate and network coherence by several orders of magnitude, compared to a naive Euler scheme without spike-time interpolation.[18]

III. SMALL-WORLD NETWORK CONNECTIVITY

Although there are many models of neuronal network connectivity,[21] including variations of random graphs[22] and scale-free networks featuring highly connected hub cells,[23] the most common is the Watts-Strogatz small-world network model.[24] This algorithm for generating network connectivity elegantly incorporates the mixture of regularity and randomness observed in many complex networks, using a single parameter to interpolate between a completely locally connected network and a random network. The model starts with a network in which each of N neurons send outgoing connections to their q nearest neighbors. Every connection is then considered for "rewiring," with the rewiring probability p used to randomly specify whether a given connection should be broken and sent to another neuron, which is itself selected randomly and uniformly from the rest of the network. Small values of p generate networks with mostly local connectivity but with a few long-range "short-cuts," and $p = 1$ generates networks in which every connection projects to a randomly selected neuron (see Fig. 2).

Watts and Strogatz showed that their model accurately reproduces the key statistical properties of many real-world complex networks (including the neuronal network of the nematode *C. elegans*, the only organism whose entire neural connectivity is completely known), thereby launching the burgeoning field of network science.[25] The field of network science applies tools from graph theory and statistical physics to characterize the structure of complex networks, which are abstracted as a set of nodes (such as neurons) and a set of connections between these nodes.

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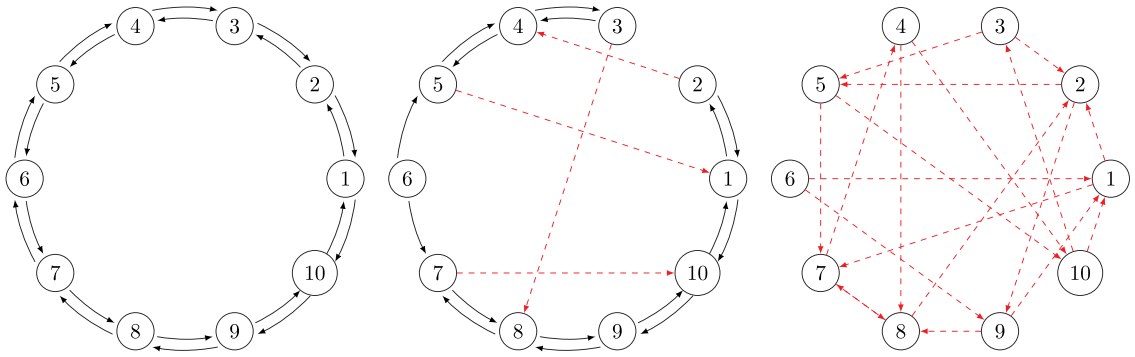

Fig. 2. Example networks generated using the Watts-Strogatz small-world network algorithm. The rewiring probability p interpolates between (a) a locally connected network and (c) a random network, with intermediate values of p generating a mixture of short- and long-range connections (b). Dashed lines indicate rewired connections.

One fundamental theoretical tool for analyzing complex networks is the adjacency matrix, an $N \times N$ matrix describing all the connections within an N-node network. If node j sends a connection to node i, then $A_{ij} = 1$; if not, $A_{ij} = 0$. For undirected networks, the adjacency matrix is by definition symmetric (though neuronal networks are most realistically modeled as directed networks, because chemical synapses are inherently directional). In a directed network, the number of incoming connections to node i (the *in-degree* of node i) may be computed as $k_i^{\text{in}} = \sum_j A_{ij}$, and the number of outgoing connections from node j (the *out-degree* of node j) is given by $k_j^{\text{out}} = \sum_i A_{ij}$. (In undirected networks, there is no distinction between in-degree and out-degree, and we refer simply to the *degree* k_i of node i.)

The adjacency matrix may also be used to compute the number of paths of length ℓ connecting any two nodes in a network. For example, if we wish to compute the number of paths of length two running from node k to node i, we must first consider all nodes that receive connections from node k, and then count how many of these send connections to node i. This algorithm may be cast in terms of matrix multiplication[26] as $\sum_j A_{ij} A_{jk} = (\mathbf{A}^2)_{ik}$. In general, $(\mathbf{A}^\ell)_{ij}$ gives the number of paths of length ℓ starting from node j and ending on node i. This clever computation is extremely useful in calculating an important network measure introduced by Watts and Strogatz known as the *clustering coefficient*. The clustering coefficient associated with a particular node is the fraction of the node's neighbors that are themselves connected to another neighbor. (In terms of social networks, it is the answer to the question, "What proportion of pairs of your friends are friends with one another?") In the Watts-Strogatz small-world network model, the clustering coefficient is a measure of the extent to which an individual node is locally connected. The clustering coefficient of node i is given by

$$C_i = \frac{\text{number of triangles centered on node } i}{\text{number of possible triangles centered on node } i}. \tag{4}$$

In an undirected network, the adjacency matrix may be used to compute C_i as

$$C_i = \frac{(\mathbf{A}^3)_{ii}/2}{k_i(k_i-1)/2} = \frac{(\mathbf{A}^3)_{ii}}{k_i(k_i-1)}. \tag{5}$$

The denominator of Eq. (5) is the binomial coefficient $\binom{k_i}{2}$, giving all the combinations of pairs of neighbors of node i. The numerator is the number of paths of length three (or triangles) that start and end at node i. The factor of two is included in the numerator to account for the fact that each triangle may be traversed in two different directions (see Fig. 3(b), as well as Appendix B for how this calculation may be extended to directed networks.) The overall clustering coefficient C of an entire network is the average of C_i over all nodes.

In the Watts-Strogatz small-world network model, the clustering coefficient is contrasted with the *characteristic path length*, which is determined by first computing the shortest path length between all pairs of nodes (see Appendix C for a description of this algorithm), and then taking the average of these values. Figure 4 depicts how the characteristic path length and clustering coefficient (each normalized to their values at $p = 0$) vary with p. The characteristic path length decreases dramatically with the introduction of just a few random connections, but in this regime the network is still mostly locally connected, resulting in the clustering coefficient declining much more gradually with increasing p. This "small-world regime," with very low

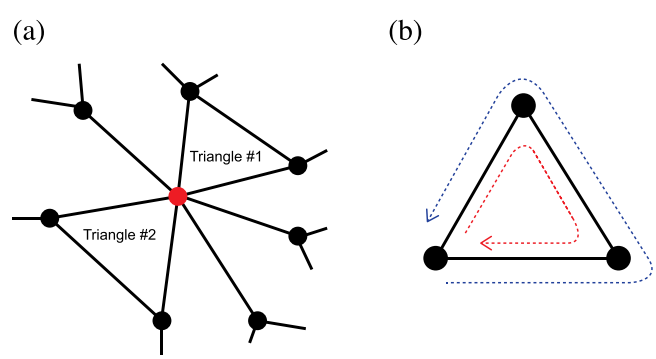

Fig. 3. Example calculation of the clustering coefficient. (a) The highlighted node has seven neighbors, and therefore, $\binom{7}{2} = 21$ possible triangles may be centered on node i. Only two such triangles exist, giving a clustering coefficient of 2/21 for the highlighted node. (b) The factor of two in the numerator of the middle of Eq. (5) is due to the fact that each triangle may be traversed in two different directions.

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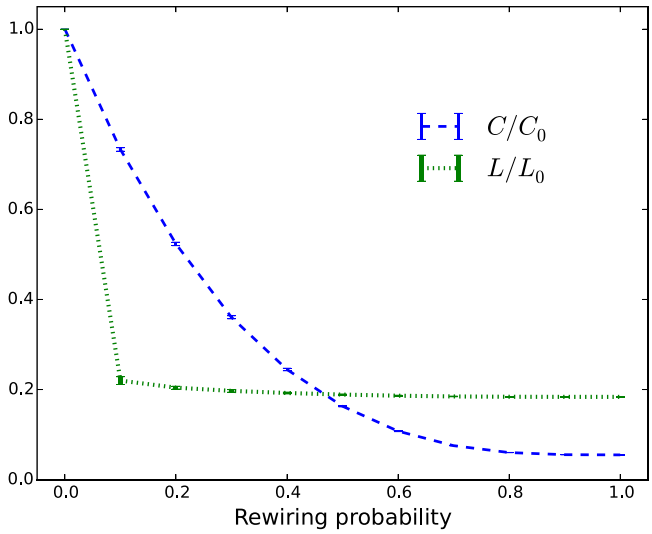

Fig. 4. The normalized characteristic path length L/L_0 and clustering coefficient C/C_0 plotted versus the rewiring probability for a network of 500 neurons (with each neuron sending connections to $q = 20$ other neurons). Note that the characteristic path length drops much faster than the clustering coefficient, leading to a small world effect for small p in which the network is mostly locally connected, yet just a few shortcut connections make the average separation between any two neurons very small. The clustering coefficient and characteristic path length were computed using the `clustering_coef_bd` and `breadthdist` functions from the Brain Connectivity Toolbox (Ref. 28).

characteristic path length but high clustering coefficient, is where many brain networks reside.[27]

IV. NETWORK SIMULATIONS

A. Simulating synaptic coupling

As we shall see, the rewiring probability heavily influences the synchronization of small-world networks,[29] including neuronal networks. To simulate the dynamics of a network of leaky integrate-and-fire neurons, we need to incorporate a model of synaptic coupling between neurons. When a neuron spikes, it influences its neighbors' membrane potentials by releasing neurotransmitter molecules that bind to proteins embedded in their cell membranes, thereby affecting currents through these ion channels.[30] From a modeling perspective, a presynaptic spike therefore results in a postsynaptic pulse in the conductance g_{syn} [see Fig. 1(a)]. This conductance pulse is often modeled as a difference between two exponential functions

$$s_{ij}(t) = \bar{g}\left[\exp\left(-\frac{t-t_{ij}}{\tau_d}\right) - \exp\left(-\frac{t-t_{ij}}{\tau_r}\right)\right], \tag{6}$$

where $s_{ij}(t)$ is the conductance pulse experienced by neurons receiving connections from neuron i due to its jth spike at time t_{ij}. The parameters τ_r and τ_d determine the rise time and decay time of the pulse, respectively, and $\bar{g}$ is the global synaptic coupling strength, applied to every connection throughout the network.

To efficiently calculate the synaptic currents received by all neurons at a given time [see Eq. (1)], we first define a vector $\vec{g}_{\text{out}}(t)$ that contains the output conductance of each neuron due to all previous spikes in time

$$g_{\text{out},i}(t) = \sum_j s_{ij}(t). \tag{7}$$

In principle, the summation runs over all spikes of neuron i that occurred previous to the current time t, but in practice, we may reduce the computational costs by omitting spikes that occur more than $\approx 5\tau_d$ previous to t, because their associated conductance pulses will have decayed to nearly zero. This approach may be implemented by maintaining a list for each neuron of only the spikes that have occurred at times later than $t - 5\tau_d$, and using this list to compute $\vec{g}_{\text{out}}(t)$ (which we take to be a column vector) at the beginning of every time step.

We refer to Eq. (1), where the synaptic current received by neuron i is given by $I_{\text{syn},i}(t) = -g_{\text{syn},i}(t)[V_i(t) - V_{\text{syn}}]$, and define the matrix $\mathbf{B(t)}$ according to $B_{ij}(t) = A_{ij}[V_{\text{syn}} - V_i(t)]$. We then compute all synaptic currents according to the matrix equation

$$\vec{I}_{\text{syn}}(t) = \mathbf{B}(\mathbf{t})\vec{g}_{\text{out}}(t). \tag{8}$$

Here, $\vec{I}_{\text{syn}}(t)$ is a column vector, with each entry giving the synaptic current received by the corresponding neuron. Casting this computation in terms of matrix multiplication allows us to take advantage of highly optimized linear algebra libraries (such as LAPACK[31]), thus boosting the computational efficiency. Once the synaptic currents have been computed at the beginning of each time step, the numerical integration scheme outlined in Sec. II may be applied to step the entire system forward in time. (PYTHON code implementing the foregoing network simulation is freely available as supplementary material.[32])

B. Measuring synchronization

To quantify the level of synchronous spiking exhibited by a simulated neuronal network, there are many measures from which to choose.[33–35] Perhaps the most straightforward approach[36] relies on the fact that in a perfectly synchronous network, the population-averaged voltage trace, $\bar{V}(t) = (1/N)\sum_{i=1}^{N} V_i(t)$ (where $V_i(t)$ denotes the voltage trace of neuron i), is equal to any one individual voltage trace, with a large "fluctuation" observed with every spike. As synchrony decreases in a population, the fluctuations in $\bar{V}(t)$ become smaller, because action potentials do not superimpose coherently. We may therefore use the time-averaged variance of $\bar{V}(t)$ as a proxy for population synchrony, with a larger variance implying greater synchrony.[37] The resulting synchronization measure χ may be normalized to the maximum value of one by dividing by the mean value of the variances of the individual voltages traces

$$\chi^2 = \frac{\sigma_{\bar{V}}^2}{\frac{1}{N}\sum_{i=1}^{N}\sigma_{V_i}^2}. \tag{9}$$

Here, $\sigma_V^2 = \langle [V(t)]^2 \rangle_t - [\langle V(t) \rangle_t]^2$, with $\langle \cdots \rangle_t$ indicating a time average. (In large-scale simulations, it may be impractical to save the complete voltage trace of every neuron. As an alternative, we may save just the time and neuronal identity number associated with every spike, then convolve each spike train with a Gaussian waveform to reconstruct approximate voltage traces off-line.)

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Table I. The parameters for the leaky integrate-and-fire model used in the network simulations (adapted from Ref. 38). In network simulations, the global synaptic coupling strength parameter $\bar{g}$ was varied, and I_{ext} was slightly different for each neuron (uniformly randomly distributed over the interval 490–510 pA).

Parameter	Value
C	0.526 nF
g_L	26.3 nS
V_{rest}	−70.0 mV
V_{syn}	0.0 mV
V_{thresh}	−52.0 mV
t_{refrac}	2.0 ms
τ_r	0.5 ms
τ_d	2.0 ms

C. Simulation results

Equation (9) was used to quantify the synchronization of simulated networks of 100 neurons with Watts-Strogatz small-world network connectivity, where each neuron sends outgoing connections to $q = 10$ other neurons. The biophysical parameters are listed in Table I, and a step size of 0.05 ms was used for the numerical integration. Figure 5(a) depicts how the synchronization varies with the rewiring probability and synaptic coupling strength $\bar{g}$. Increases in either parameter result in greater network synchrony. Note how the introduction of just a few random connections dramatically influences the network dynamics.

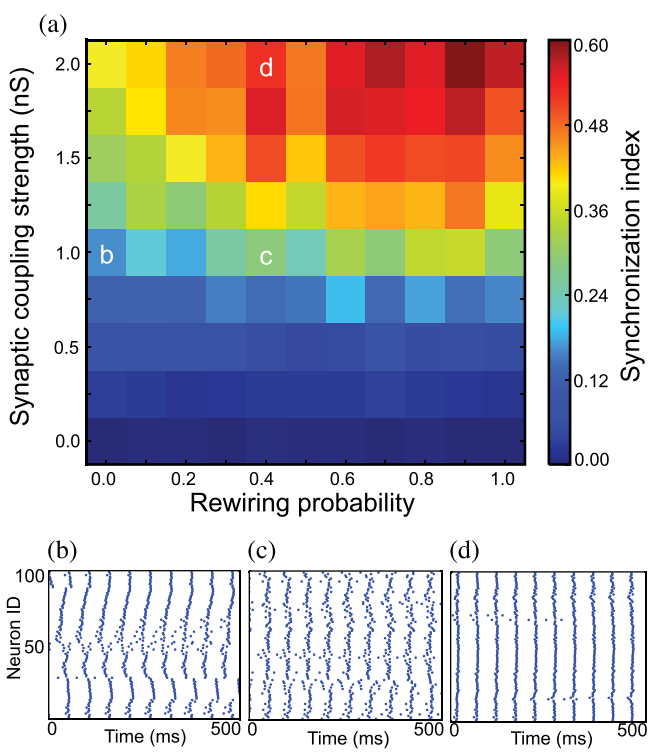

Fig. 5. Synchronization of a network of leaky integrate-and-fire neurons with Watts-Strogatz small-world network connectivity. (a) The synchronization index χ as a function of the rewiring probability and coupling strength. Each square represents the average value of χ over ten simulations. (b)–(d) Raster plots depicting neuronal spike timing for the network parameters indicated by labeled squares in (a). Note the increase of the synchronization as the rewiring probability increases from zero in (b) to a non-zero value in (c), and how synchronization increases with greater coupling strength from (c) to (d). A vertical line indicates perfect synchronization.

Figures 5(b)–5(d) are raster plots of network activity, in which a spike by neuron i at time t is represented as a dot at the corresponding coordinates (t, i). Vertical bands in the raster plot therefore indicate large numbers of neurons firing at similar times, that is, synchrony. Figures 5(b) and 5(c) show the effect of increasing just the rewiring probability, while Figs. 5(c) and 5(d) show the effect of increasing just the coupling strength, underscoring the influence that both parameters exert on network synchronization.

V. CONCLUSIONS

Although the fact that increased coupling strength enhances synchrony may be reasonably intuitive, the effect of increased rewiring probability is not as obvious. The observation that this effect also enhances synchrony may be heuristically explained by the fact that as p increases, the average shortest path length decreases, which enhances information transmission within a network, thereby enhancing synchrony.[39] (This explanation is an oversimplification, as several other structural characteristics of networks have also been shown to influence synchronizability,[40,41] with the master stability function analysis having the final say in the matter.[42]) One application of this result is that in the brain, the enhanced synchrony resulting from rewired connections may explain the emergence of epilepsy following pathological axonal sprouting, in which new connections are generated following a brain injury.[43]

We emphasize that the network model we have discussed is quite basic and can be made more biophysically realistic in several ways. Synaptic conductances can be modeled using differential equations to enforce a saturation limit,[44] for instance, and a more realistic neuron model (based on the Hodgkin-Huxley formalism[17]) can be used. Most importantly, inhibitory cells (which inhibit the firing of other neurons, and may be modeled simply by changing the associated values of V_{syn} from 0 mV to −80 mV) could be incorporated within the network,[38] because approximately 20% of neocortical neurons are inhibitory.[45]

ACKNOWLEDGMENTS

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APPENDIX A: SUGGESTED PROBLEMS

1. Derive the reset potential for the leaky integrate-and-fire neuron, Eq. (3).
2. Write a program to generate the adjacency matrix for a directed small-world network with number of nodes N, rewiring probability p, and out-degree q. Use your program in conjunction with the functions `clustering_coef_bd` and `breadthdist` from the Brain Connectivity Toolbox[28] to reproduce Fig. 4.
3. Write a program to simulate a network of 100 neurons with small-world network connectivity, as described in Sec. IV. Then modify your simulation to include 20 inhibitory cells by setting $V_{syn} = -80$ mV for all inhibitory

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connections. Also, modify your small-world network connectivity algorithm so that when an inhibitory connection is rewired, it projects to an excitatory neuron with probability p_{xi} (and to an inhibitory neuron with probability $1 - p_{\mathrm{xi}}$). Likewise, when an excitatory connection is rewired, it should project to an inhibitory neuron with probability p_{ix} (and to an excitatory neuron with probability $1 - p_{\mathrm{ix}}$). Explore the varieties of network dynamics that result from various combinations of parameters. One particularly important dynamical mode is the *gamma oscillation*,[46] which is an oscillation in the local field potential (recorded by positioning an electrode outside of a population of neurons, thereby obtaining spatially averaged neural activity) of 40–80 Hz. The local field potential (LFP) may be modeled as the total synaptic current [from Eq. (8)], summed over all neurons: $\mathrm{LFP}(t) = \sum_i I_{\mathrm{syn},i}(t)$. For what parameters do gamma oscillations emerge?

APPENDIX B: DIRECTED CLUSTERING COEFFICIENT

To extend the clustering coefficient to directed networks,[47] we may start with Eq. (4), but recognize that there are now eight different "triangles" that may surround a particular node [see Fig. 6(a)]. These additional triangles may be taken into account by defining the symmetrized adjacency matrix, $S = A + A^T$, for which S_{ij} gives the total number of connections (0, 1, or 2) in either direction between nodes i and j. The total number of triangles (of any of the eight varieties) surrounding node i is $(S^3)_{ii}/2$, analogous to the undirected case described in Eq. (5). To compute the total number of pairs of neighbors of node i, we first define the total degree as $\beta_i = k_i^{\mathrm{in}} + k_i^{\mathrm{out}} = \sum_j S_{ij}$.

In general, the total number of possible triangles will be less than $\binom{\beta_i}{2}$ due to paths of length two that start at node i, project to another node, and then return to node i [see Fig. 6(b)]. Because such paths cannot possibly generate a triangle, they must be subtracted from $\binom{\beta_i}{2}$, giving $\binom{\beta_i}{2} - (A^2)_{ii}$. Finally, each of the remaining pairs may generate two triangles, because the two nodes may themselves be connected in two different directions. Altogether, we may define the clustering coefficient of node i in a directed network as

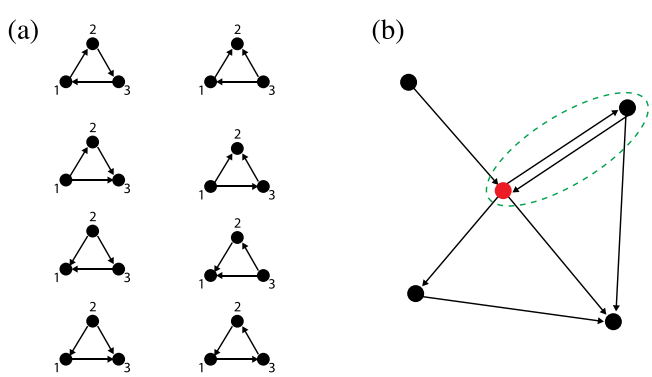

Fig. 6. Calculation of the clustering coefficient in directed networks. (a) In contrast to undirected networks, directed networks permit eight varieties of triangles to be centered on a particular node. (b) In counting the total number of possible triangles centered on a particular node, reciprocal connections such as those indicated in the diagram must be subtracted from $\binom{\beta_i}{2}$.

$$C_i = \frac{(S^3)_{ii}/2}{2\left[\beta_i(\beta_i - 1)/2 - (A^2)_{ii}\right]} = \frac{(S^3)_{ii}}{2\left[\beta_i(\beta_i - 1) - 2(A^2)_{ii}\right]}. \tag{B1}$$

The function `clustering_coef_bd` from the Brain Connectivity Toolbox[28] computes the clustering coefficients for all nodes in a directed network.

APPENDIX C: SHORTEST PATH LENGTHS

The *breadth-first search* algorithm[26] is used to compute the shortest paths from a given node i to all other nodes in a network. The idea is simple: start with node i, which is by definition a distance 0 from itself. Find all the neighbors to which it sends connections (using the adjacency matrix, for example), which are recorded as being a distance one from node i. Then find all the neighbors of those neighbors (which have not already been considered), assign them a distance of two, and continue this process of finding neighbors of neighbors, incrementing the distance by one with each new round. By keeping a list of all the nodes in the most recent round and using it to determine the neighbors in the next round, this algorithm takes $\mathcal{O}(N)$ time (where N is the number of nodes) for most real-world networks.[26] To compute the characteristic path length of an entire network, run a breadth-first search for all nodes in a network, thereby constructing a matrix $\mathbf{d}$ that stores the lengths of the shortest paths between all pairs of nodes. (Note that for a directed network, this matrix will not in general be symmetric.) The characteristic path length is then given by $1/(N(N-1))\sum_{i \neq j} d_{ij}$. The function `breadthdist` from the Brain Connectivity Toolbox[28] generates this distance matrix.

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